

AI Course – Revision Notes (Lessons 1–7)

Lesson 1 – What is Artificial Intelligence (AI)?

Section 1 – Main Ideas

Big Picture – What is AI?

- AI = artificial (made by humans) + intelligence (learning, solving problems, using what you know to figure things out).
- AI is about building machines that can perceive, understand, reason, learn and act intelligently; they don't have to think like humans, only behave intelligently.
- AI works by finding patterns in data and using them to predict what comes next, so it can do some mental tasks better than humans, but it still cannot feel or truly understand.

Humans, Technology and Change

- Technology is anything humans create to make life easier (taps, cars, umbrellas), and new technologies usually replace older ways of doing things.
- AI is “thinking technology” that helps with remembering information, repetitive thinking, following instructions, language translation, basic writing, maths and summarising.
- Like cars replacing walking only for long distances, AI replaces certain types of thinking but not all; humans are still better at deep context, moral judgement and dealing with totally new problems.

Early AI – Rules and Limits

- In 1950 Alan Turing proposed the Turing Test: if you talk to a machine and cannot tell it from a human, it can be considered intelligent.
- In the 1950s–60s early AI was rule-based: scientists tried to turn intelligence into lots of “if this, then that” rules, creating systems like the perceptron and ELIZA.

- Rule-based AI could only handle narrow tasks and needed huge numbers of hand-coded rules, so it could not adapt to real life, leading to disappointment and an “AI winter” in the 1970s.

Machine Learning and Deep Learning

- Machine learning changed AI from being told exactly what to do into learning from data, like teaching a baby by showing examples instead of giving step-by-step rules.
- Three learning styles are: supervised (labelled examples), unsupervised (finding patterns without labels) and reinforcement learning (trial and error with “right/wrong” feedback).
- With more data and the backpropagation algorithm, neural networks (layers of artificial “neurons”) could adjust themselves after mistakes and slowly learn complex tasks.
- Deep learning stacks many layers so the AI builds simple patterns into complex ones (edges → shapes → objects → scenes), powering tools like voice assistants, translation, face ID and autocorrect.

Transformers and Generative AI

- Around 2017 transformers were invented to fix a key limit: older systems could only focus on short bits of text and would “forget” what came earlier.
- Transformers use attention to look at many words at once and track how they relate (for example understanding that “he” refers back to “the child”), which made large language models like ChatGPT possible.
- Generative AI models such as GPT predict the next token again and again, allowing them to generate long text, code, images or video that often seems human but is really a clever remix of learned patterns.
- Multimodal AI is like one “super baby” trained to read, see, listen and talk with a single model, but it still does not feel or truly know anything, so it can make strange or wrong outputs.

Why AI Is Booming and Why It Matters

- Modern AI exploded when four ingredients combined: lots of data, powerful computers, big investment and breakthroughs like deep learning, reinforcement learning and transformers.

- Giving AI a body with sensors and motors creates physical AI such as robots; other branches like robotics and computer vision still share the same goal of building intelligent machines.
- We already use AI every day through recommendations, autocorrect and photo filters, and we are living through a shift from “pre-AI” to “post-AI”, so understanding AI helps us keep up as it changes life and work.

Section 2 – Key Terms and Definitions

- Artificial Intelligence (AI) – The science of building machines that can perceive, understand, reason, learn and act intelligently.
- Technology – Anything humans create to make life easier, solve problems or achieve goals, from taps to cars to AI systems.
- Turing Test – A test where you chat with a machine; if you cannot tell it from a human, the machine is said to be intelligent.
- Rule-based AI – Early AI that followed hand-written “if this, then that” rules to make decisions, with little ability to adapt.
- Machine Learning – A way for computers to learn from data instead of following only fixed rules programmed by humans.
- Supervised Learning – Machine learning that uses labelled examples (input plus correct answer) to train a model.
- Unsupervised Learning – Machine learning that looks for patterns and groups in data without any labels.
- Reinforcement Learning – Learning by trial and error, where an AI is rewarded for good actions and penalised for bad ones.
- Neural Network – A collection of connected artificial “neurons” that can learn patterns from data.
- Backpropagation – An algorithm that lets neural networks adjust their internal connections when they make mistakes, so they improve over time.
- Deep Learning – Machine learning using neural networks with many layers to learn complex, multi-level patterns.

- Transformer – A neural network architecture that uses attention to handle long sequences of data, especially text.
- Attention (in AI) – A mechanism that lets a model focus on the most relevant parts of its input, such as linking “he” back to “the child” in a sentence.
- Generative AI – AI systems that can create new content (text, images, code, video) by predicting the next part of a pattern.
- Large Language Model (LLM) – A generative AI model trained on huge amounts of text to predict and generate language.
- Token – The small unit of text used by language models (a word or part of a word) that they predict one after another.
- Multimodal AI – An AI model that can handle more than one type of input or output, such as text, images and audio, using a single “brain”.
- Pattern Recognition – The ability of AI to spot regularities in data, such as shapes in images or structures in sentences.
- Physical AI / Robotics – AI systems given a body with sensors and motors so they can move and act in the physical world (robots).
- Datafication – The process of turning many parts of life into digital data that AI can use for learning.

Section 3 – Short Summary

This lesson explains what artificial intelligence is and why it marks a major shift from a “pre-AI” to a “post-AI” world. AI means building machines that can learn, reason and act intelligently by recognising patterns in data and making predictions. Early AI relied on hand-written rules and quickly hit limits, which led to a new focus on machine learning and deep learning, where models learn from examples instead of fixed instructions. Transformers then allowed AI to handle long, complex text using attention, powering large language models and today’s generative AI systems. These models can now create text, images, code and more that often look human, even though they are really remixing learned patterns and do not truly understand or feel. Modern AI became powerful when huge amounts of data, strong computers, investment and key breakthroughs all came together. Because AI already affects how we learn, work and relax, understanding how it works helps us use it wisely instead of being left behind.

Section 4 – Quick Check Questions

1. Define artificial intelligence (AI) in your own words.
 2. Explain the main difference between rule-based AI and machine learning.
 3. Give one everyday example of deep learning in action.
 4. Why did transformer models make such a big difference to what AI can do with language?
 5. Name two reasons why modern AI has become so powerful in recent years.
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Lesson 2 – Today’s AI: From Chatbots to Agents

Section 1 – Main Ideas

From Gen AI to Agents

- Earlier AI mostly analysed and organised information; generative AI added “make this for me” by creating text, images, music and code.
- The next step is “handle this for me” – AI systems that can not only create content but also plan and carry out tasks, like a real assistant.
- This shift is like moving from practising exam questions in class to independently sitting the exam.

How ChatGPT Generates Answers

- Your message is broken into tokens (small chunks of text), which are turned into numbers so the AI can do maths on them.
- The model processes all tokens in its context window (short-term memory), covering the whole conversation until that window fills up.
- It predicts the most likely next token, adds it, then predicts the next one, repeating until it eventually predicts a “stop” token and ends the answer.

Training, Fine-Tuning and RLHF

- Training (pre-training) happens once: the base model learns patterns from huge amounts of text and other data; using it in chat is cheap and does not change its core knowledge.
- A base model (like raw GPT) knows patterns in language but has not been trained to be helpful or safe.
- Fine-tuning turns a base model into an assistant model (like ChatGPT) by teaching it how to behave, often using RLHF – humans rate answers so the AI learns which responses are good or bad.

Autonomy Ladder – LLMs, Workflows, Agents

- Level 1 LLMs: you ask, it answers; it only generates outputs and cannot take actions like emailing or saving files.
- Level 2 workflows: a fixed sequence of steps is set up in advance, so the AI can act (e.g. format, save, email an essay) but only by following your pre-written script.
- Level 3 agents: you give a goal (“Plan my week”) and the AI decides how to achieve it, choosing tools and actions itself, but still needing supervision.
- This “autonomy ladder” is like growing from a child (LLM) to a teen (workflow) to a more independent helper (agent).

Inside an AI Agent

- Brain: an LLM that understands instructions, reasons with information and chooses what to do next.
- Memory: short-term (context window in the current conversation) plus long-term memory stored in a database across sessions.
- Tools: ways to act in the world, such as searching the web, sending emails, reading/writing files, using calendars or generating images.
- Prompt and knowledge: a system prompt gives the agent its “job description”, and extra knowledge (like your timetable or deadlines) lets it personalise its decisions.

Evaluating AI Systems – 5 Questions

- What patterns did it learn from (text, images, code)? → Tells you what it's good or bad at.
- Is it just generating, or can it take action? → Tells you if you must do the rest of the work yourself.
- What does it remember, and for how long? → Tells you how focused each conversation should be.
- What tools does it have access to, and who is making the decisions (you or the AI)? → Tells you what's possible and how much you need to supervise it.

Section 2 – Key Terms and Definitions

- Token – A small chunk of text (a word or part of a word) that an AI model processes and predicts step by step.
- Context Window – The amount of recent text (tokens) the AI can “keep in mind” at once, like its short-term memory.
- Training / Pre-Training – The one-time, expensive process where a base model learns patterns from massive datasets.
- Using (Inference) – Running the trained model to answer your prompts; it is fast, cheap and does not normally update the base model.
- Base Model – A model that has learned general patterns from data but has not yet been shaped to be a helpful assistant.
- Fine-Tuning – Extra training that turns a base model into a specialised assistant model with a particular behaviour.
- Assistant Model – A fine-tuned model (like ChatGPT) designed to be helpful, safe and user-friendly.
- RLHF (Reinforcement Learning from Human Feedback) – A training method where humans rate AI answers so the model learns which responses are better.
- LLM (Large Language Model) – A model trained on huge amounts of text that generates language by predicting the next token.

- Workflow – A predefined series of steps an AI system follows to complete a task, without making its own high-level decisions.
- Agent – An AI system that can decide which actions and tools to use to reach a goal, not just generate text.
- Autonomy Ladder – The idea that AI systems range from simple LLMs to workflows to agents, gaining more independence at each level.
- Short-Term Memory – Information the AI can use within the current context window of a conversation.
- Long-Term Memory – Information stored outside the context window (e.g. in a database) that an agent can refer back to across sessions.
- Tools (in AI) – External actions an agent can perform, such as web search, file editing, emailing or accessing calendars.
- System Prompt – A hidden instruction that tells the AI its role and how it should behave (its “job description”).
- Knowledge (for Agents) – Extra, often personal or organisational information an agent uses, like schedules, deadlines or policies.
- AI Evaluation Questions – The five questions (patterns, actions, memory, tools, decision-maker) used to quickly understand any AI system.
- Privacy (in Agents) – The concern about what personal data an agent can access (emails, calendars, files) and how safely it uses that data.

Section 3 – Short Summary

This lesson explains how today’s AI systems work, from simple chatbots to more independent agents. When you talk to a model like ChatGPT, your words are broken into tokens, processed within a limited context window, and turned into answers by predicting one token at a time. The model was trained once on huge datasets (pre-training), then fine-tuned with methods like RLHF so it behaves as a helpful assistant rather than just completing text. Modern AI is climbing an autonomy ladder: LLMs only generate text, workflows follow fixed multi-step recipes, and agents can choose tools and actions to reach a goal. An agent combines a brain (LLM), memory, tools, a system prompt and extra knowledge to manage tasks like planning your homework schedule. To understand any AI tool, you can ask five key questions about its training data, actions, memory, tools and who makes the decisions. These fundamentals help you judge what AI can and cannot do, even as technology continues to advance.

Section 4 – Quick Check Questions

1. What is a context window, and why can an AI “lose track” in very long conversations?
 2. Explain the difference between a base model and a fine-tuned assistant model.
 3. Describe one key difference between an LLM, a workflow and an agent.
 4. Name and briefly explain two parts of an AI agent’s “architecture” (inner design).
 5. Why is it important to think about tools, memory and decision-making when you evaluate how much to trust an AI system?
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Lesson 3 – The AI Market: From Lab to Your Phone

Section 1 – Main Ideas

From Model to Product

- AI goes from research lab to your phone in a chain: Lab → Model → API → Product → You.
- Big companies, startups and universities spend huge money and time training models – this “engine” is powerful but too technical for normal users.
- An API (Application Programming Interface) is like a waiter between you and the “kitchen” model: it takes your request (prompt), sends it to the model, and returns the result.
- Product builders use the API to add interfaces (chat boxes, buttons, voice) and features, turning one engine into many different apps like ChatGPT, Copilot or Notion AI.

The AI Pyramid – Who Does What?

- The pyramid metaphor shows a few model makers at the top, many product builders in the middle, and millions of users at the bottom.
- Model makers (OpenAI, Google, Anthropic, Meta) focus on improving core AI engines; product companies (Grammarly, Canva, Jasper, Notion) focus on specific user needs.

- You never “touch” the raw model; you only interact with the product built on top of it to get your work done.

Four Types of AI Products (What + Where)

- Two key questions: Is it a *generalist* (does many tasks) or *specialist* (one task very well)? And is it *standalone* (separate app) or *integrated* (built into tools you already use)?
- Type 1 – Generalist Standalone: versatile apps like ChatGPT, Claude or Gemini; like an SUV that handles many situations.
- Type 2 – Specialist Standalone: single-skill apps like Midjourney (images), ElevenLabs (voices) or Perplexity (search); like a sports car or ambulance built for a specific job.
- Type 3 – Generalist Integrated and Type 4 – Specialist Integrated live inside other software (e.g. Copilot in Office, Grammarly in your browser), like built-in GPS or a backup camera in a car.

Who Controls What? (Control Levels 1–4)

- There is a spectrum of control from “they control everything” to “you control everything”.
- Level 1: They control everything (most consumer apps); you only choose prompts – like taking a taxi. Easy, but not very customisable.
- Level 2: They control, you edit; you adjust settings and style, but they run the model – like renting a car with options.
- Levels 3–4: You design behaviour, tools and even the model and hardware; this gives maximum customisation and privacy but needs time, skills and good equipment.

Autonomy Levels and Product Types

- LLMs usually appear as Type 1 or Type 3 products: generalist helpers either as separate apps or built into tools.
- Workflows usually show up as Type 4 specialist features, like very specific AI add-ons that follow fixed steps.
- Agents can be any type but are increasingly built as powerful specialists (often Type 2), like a self-driving car that chooses how to reach your goal.

Costs, Business Models and Access

- Training big models is extremely expensive (millions to hundreds of millions of dollars), but using them (inference) is very cheap per request.
- Many tools are “free” because of business models like freemium (basic free, advanced paid), enterprise sales (business pays) or loss leaders (free now, money later) – sometimes also by using your data.
- Access to powerful AI can depend on money: unlike a luxury car you don’t *need*, missing out on key AI tools could become a real disadvantage for study and work.

AI, Jobs and Preparing for the Future

- The car example shows that new tech destroys some jobs (blacksmiths, carriage makers) but creates others (mechanics, drivers, road builders).
- Smart workers shifted from horses to cars; in the same way, we can shift from purely human mental work to roles that use and guide AI.
- AI is different from cars because it replaces parts of *intelligence*, not just physical effort, so the shift is bigger – understanding AI early helps you prepare instead of being left behind.

Section 2 – Key Terms and Definitions

- Model – The core AI “engine” that has been trained on large datasets to recognise patterns and generate outputs.
- API (Application Programming Interface) – A connection that lets one program ask another (the model) to do a task and return the result, like a waiter carrying orders to a kitchen.
- AI Product – An app or service built on top of a model and API, with an interface and features for normal users (e.g. ChatGPT, Copilot).
- AI Pyramid – A picture of the market: a few model makers at the top, many product builders in the middle, and millions of users at the bottom.
- Generalist AI – A tool that can handle many different tasks such as writing, analysis and coding (e.g. ChatGPT).

- Specialist AI – A tool focused on one main task, such as image generation or grammar checking (e.g. Midjourney, Grammarly).
- Standalone – An AI you visit as a separate app or website, like opening ChatGPT or Midjourney directly.
- Integrated – AI that is built into software you already use, like AI inside Google Docs or Microsoft Word.
- Type 1–4 Products – The four combinations of generalist/specialist and standalone/integrated that describe most AI tools.
- Control Levels 1–4 – A spectrum showing who controls the AI settings and behaviour, from company-controlled (Level 1) to fully user-controlled (Level 4).
- Inference – The process of *using* a trained model to answer prompts; cheap per use compared to training costs.
- Freemium Model – A pricing strategy where basic use of an AI tool is free but advanced features require payment.
- Enterprise Sales – When companies sell AI tools to organisations or businesses rather than individual users.
- Loss Leader – A product offered cheaply or for free at first to gain users, with plans to make money later.
- Market Competition – Rivalry between AI companies (like different phone or console brands) that speeds up progress and constant new releases.
- Job Displacement – When technology reduces or removes certain jobs, such as carriages being replaced by cars.
- Reskilling – Learning new skills so workers can move into new roles created by technology, like blacksmiths learning to work with cars.
- Mental vs Physical Work – Physical work uses the body (lifting, building); mental work uses thinking and problem-solving, which AI is increasingly able to help with.

Section 3 – Short Summary

This lesson explains how AI moves from research labs into the apps on your phone and how to make sense of the crowded AI market. First, powerful models are trained, then exposed through

APIs so companies can build user-friendly products on top; a small group of model makers supports many product builders and millions of users. You can understand any AI tool using two questions: is it a generalist or specialist, and is it standalone or integrated, giving four main product types. A second lens is control: from simple tools where the company controls everything to advanced setups where you control models, data and workflows. The lesson also covers the high cost of training models, why many tools seem “free”, and how access to AI may depend on money. Finally, using the car analogy, it shows how new technologies like AI both remove and create jobs, and that learning to work *with* AI is key to thriving in an AI-shaped future.

Section 4 – Quick Check Questions

1. Describe the journey from AI model in a lab to the app you use on your phone.
 2. What are the two questions that define the four main types of AI products, and give one example type.
 3. Explain one of the control levels (1–4) using the car analogies from the lesson.
 4. Why is training AI much more expensive than using it, and how does this affect business models like “free” tools?
 5. How is the impact of AI on jobs similar to, and different from, the impact of cars on horse-based jobs?
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Lesson 4 – When to Use AI (and When Not To)

Section 1 – Main Ideas

AI as Superpower or Downfall

- AI is a double-edged sword: it can make you 10× more powerful, or make you weak and dependent if you use it badly.
- Like calculators in primary school or eating only bananas, using AI too early or for everything can damage your basic skills and mental “diet”.
- The key motto of the lesson is: **“Use AI – don’t let AI use you.”**

Patterns, Head Work and AI

- AI is a pattern machine: it recognises patterns in data (books, code, images) and predicts what fits the pattern next.
- Wherever work is mechanical, rule-based and follows clear patterns, AI can usually do it faster and more accurately than humans.
- This kind of pattern-based thinking is called **HEAD work** – it's where AI should mostly be used.

Heart Work – Protect It

- Some tasks have no clear pattern: they rely on judgement, emotion, values and truly original ideas – this is **HEART work**.
- HEART work includes your unique opinions, moral choices, relationships, personal growth and creative voice; there is no single “right answer”.
- AI is **terrible** at HEART work, so you must keep these tasks for yourself to protect your identity and long-term skills.

Finding the Real Goal

- Every task has a **surface goal** (e.g. “get 10/10 on homework”) and a **real goal** (e.g. “learn problem-solving for exams”).
- If you only care about the surface goal, you will overuse AI, finish the task, but learn almost nothing.
- To find the real goal, ask: “Who benefits if I do this well?”, “What am I supposed to learn or build?”, and “Is this about the output or the process?”

Breaking Tasks into Parts

- Most real tasks are a **mix** of head and heart, so you should split them into components instead of asking “AI or no AI?” for the whole thing.
- Example: in an essay, AI can help with research, grammar and citations (HEAD), while you handle arguments, ideas and conclusions (HEART).

Common AI Traps and How to Avoid Them

- **Hallucination trap:** AI confidently makes things up; you must fact-check and stay responsible for accuracy.
- **Trigger-happy trap:** reaching for AI automatically so you stop thinking for yourself, even on easy problems.
- **Automation-of-joy & invisible decay traps:** letting AI do activities you actually enjoy or over-relying on it until your skills quietly weaken.

Habits for Healthy AI Use and the Future

- Use **deliberate delay** (brief pause to ask “head or heart?”), **AI-free periods**, self-tests and personal rules to stay disciplined.
- Always give AI enough context; weak prompts lead to generic answers (the **context-free trap**).
- As AI gets better at HEAD work, pattern-based tasks become cheaper and less valuable, so strong HEART skills and adaptability become even more important.

Section 2 – Key Terms and Definitions

- **Pattern Machine** – A way of describing AI: it recognises patterns in data and predicts what usually comes next.
- **Head Work** – Mechanical, pattern-based tasks with clear steps and right answers (e.g. grammar checks, formatting, calculations) where AI excels.
- **Heart Work** – Personal, emotional and value-based tasks (e.g. big decisions, original ideas, relationships) that build who you are and should not be given to AI.
- **Surface Goal** – The obvious task aim, like “finish the worksheet” or “hand in 10 pages by Friday”.
- **Real Goal** – The deeper purpose, such as building skills, understanding or character, which should guide how you use AI.
- **Hallucination (AI)** – When an AI system produces information that looks correct but is actually made up or wrong.
- **Hallucination Trap** – Relying on AI’s fake but confident answers without checking, then making decisions based on false information.

- Trigger-Happy Trap – Automatically using AI for everything, so your first instinct is “ask AI” instead of “think”.
- Automation-of-Joy Trap – Letting AI do activities you love, so you stop practising them and lose enjoyment and personal growth.
- Invisible Decay Trap – Your skills slowly weakening because AI keeps doing the hard parts for you and you don’t notice.
- Context-Free Trap – Getting vague, generic answers because you didn’t give AI enough background or detail in your prompt.
- Deliberate Delay – A short pause before using AI to ask if it’s really needed and whether the task is head or heart.
- AI-Free Periods – Planned times or types of work where you refuse to use AI to keep your own skills active.
- Weekly Self-Test – A regular exercise where you redo a task without AI to check if your abilities are staying sharp.
- Personal AI Rules – Written guidelines you set for yourself (e.g. “no AI on first drafts”) to avoid over-reliance.
- “Use AI, Don’t Let AI Use You” – The lesson motto: AI should be a tool that boosts you, not something that quietly replaces your thinking.

Section 3 – Short Summary

This lesson explains how to decide **when** to use AI and when to rely on yourself. AI is a powerful pattern machine, great at HEAD work like grammar, formatting, research and calculations, but bad at HEART work that involves your values, emotions, identity and truly original ideas. To use AI wisely you must look beyond the surface goal of a task and identify the real goal, such as learning, growth or building judgement. Most tasks mix head and heart, so you should break them into parts and choose which pieces AI helps with and which you must do yourself. The lesson also warns about traps like hallucinations, trigger-happy use, automating things you enjoy, invisible skill decay and giving AI too little context. Finally, it offers protection tactics—deliberate delay, AI-free periods, weekly self-tests and personal rules—so AI becomes a superpower that boosts your abilities instead of slowly replacing them.

Section 4 – Quick Check Questions

1. Explain the difference between HEAD work and HEART work, and give one example of each.
 2. What is the **real goal** of homework or essays, and how should that change the way you use AI on them?
 3. Describe one AI trap (e.g. hallucination, trigger-happy, automation-of-joy or invisible decay) and why it is dangerous.
 4. What is one personal AI rule you could create to avoid over-relying on AI?
 5. Why are HEART skills likely to become more valuable as AI gets better at HEAD work?
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Lesson 5 – Speaking AI: How to Talk to Your Assistant

Section 1 – Main Ideas

Why Communication Matters

- Everyone in your class has access to the same AI; what makes the difference is *how* you communicate with it, not which tool you use.
- Most people write vague, lazy prompts and then blame the AI for vague, lazy answers.
- Think of AI as a super-capable assistant who knows a lot, but knows nothing about *you* unless you explain your situation clearly.
- Core rule: **Garbage in → garbage out, gold in → gold out** – input quality drives output quality.

Core Prompting Principles

- Specific prompts beat vague ones: adding length, topic, audience, tone and structure can turn a weak answer into a great one.
- The more the AI knows about your goal, audience, constraints and background, the better it can tailor its response.

- Good prompting is common sense if you imagine explaining the task to a human assistant who can't read your mind.

The CO-STAR Framework

- CO-STAR = **Context, Objective, Style, Tone, Audience, Response** – a structure for clear, powerful prompts.
- Context sets the scene (what you're doing, why, stage, constraints) and is the *most powerful* part because it turns generic AI into "your" AI.
- Objective tells AI what success looks like; Style and Tone shape how it sounds; Audience makes it suitable for the right people.
- Response tells AI the format you want (outline, bullets, script, table), so you don't get a useless wall of text.

Improving and Refining Prompts

- Pros treat prompting as a process: write a prompt, check the output, spot what's wrong, refine, repeat.
- When an answer isn't right, diagnose the issue (length, tone, detail, format, missing info) and edit *that* part of the prompt.
- Meta-prompting = asking AI how to improve your prompt; it can point out missing details, unclear instructions and better structure.

Advanced Prompting Tactics

- Role prompting: "Act as a [specific role] with [experience] in [area]" helps AI use the right knowledge patterns (e.g. physics teacher, editor, admissions officer).
- Few-shot prompting: show examples of the style or format you want so AI can copy the pattern ("show, don't just tell").
- Chain-of-thought and self-consistency: ask AI to think step-by-step and give multiple approaches with pros/cons before recommending a choice.

Constraints and Common Mistakes

- Smart constraints (word limits, required elements, things to avoid) push AI towards more original and useful outputs – too much freedom leads to generic answers.
- Big mistakes include: changing the question instead of refining, assuming AI “knows what you mean”, hiding your real constraints, and using AI for HEART work it’s bad at.
- The real skill isn’t “being good with AI”, it’s being a clear communicator; the same skills improve how you talk to humans too.

Section 2 – Key Terms and Definitions

- Prompt – The instruction or message you give to an AI telling it what you want it to do.
- Garbage In, Garbage Out – Idea that poor prompts produce poor answers; high-quality prompts produce high-quality outputs.
- Context – Background information about your situation, goal, constraints and level that helps AI give relevant, tailored responses.
- CO-STAR – A prompting framework: Context, Objective, Style, Tone, Audience, Response.
- Objective – What you actually want to achieve with the AI (e.g. outline, explanation, plan), not just the surface request.
- Style – How the writing should *sound*, such as conversational, academic, direct or storytelling.
- Tone – The mood or feeling of the response (e.g. serious, humorous, urgent, encouraging).
- Audience – The group the output is for (e.g. 14-year-olds, teachers, busy parents), which affects level, examples and wording.
- Response Format – The structure you ask for, like bullets, outline, table, script or essay.
- Specificity Principle – The idea that more specific prompts lead to more tailored and useful outputs.
- Meta-Prompting – Asking AI to help you improve your own prompt by analysing what’s missing or unclear.

- Output Control – Setting rules for format, length, tone, style, required elements and constraints so AI “gets it right first time”.
- Role Prompting – Telling AI to “act as” a specific expert role so it uses more appropriate knowledge and style.
- Few-Shot Prompting – Giving one or more examples of what you want so the AI can copy the style, structure or tone.
- Chain-of-Thought Prompting – Asking AI to think step-by-step and show its reasoning before giving a final answer.
- Self-Consistency – Asking AI for multiple different approaches to a problem, then comparing them to choose the best one.
- Constraint Paradox – The idea that adding smart limits (length, required elements, twists) can make AI’s answers *more* creative, not less.
- Knowledge Bank – A collection of your best prompts and expert instructions that you can reuse and adapt over time.
- Head Work – Pattern-based tasks where AI is strong (from earlier lessons), which good prompting helps it do even better.
- Heart Work – Personal, value-based tasks where AI is weak, reminding you not to hand over everything just because prompts are powerful.

Section 3 – Short Summary

This lesson teaches you how to *talk* to AI so it becomes a powerful assistant instead of a generic answer machine. The key idea is that output quality depends on input quality: vague prompts get vague results, while specific, well-structured prompts unlock AI’s real abilities. The CO-STAR framework (Context, Objective, Style, Tone, Audience, Response) helps you give AI all the information it needs, especially rich context about your situation and constraints. Prompting should be treated as a process where you test an output, spot what’s wrong, refine the prompt and repeat, using tools like meta-prompting and clear output control. Advanced tactics such as role prompting, few-shot prompting, chain-of-thought and self-consistency give you even more control and better thinking from the AI. Finally, the lesson warns about common mistakes—like changing the question instead of refining, hiding constraints and using AI for HEART work—and encourages you to build a knowledge bank of strong prompts you can reuse.

Section 4 – Quick Check Questions

1. Why does “garbage in → garbage out” matter so much when you’re writing prompts for AI?
 2. Write your own CO-STAR prompt for a homework task you actually have (just outline the six parts).
 3. Explain what meta-prompting is and how it can help you fix a weak prompt.
 4. Describe one advanced tactic (role prompting, few-shot, chain-of-thought or self-consistency) and give a simple example of when you’d use it.
 5. What is the constraint paradox, and why can adding limits sometimes make AI’s answers *better* rather than worse?
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Lesson 6 – How to Use AI Like the Top 1%

Section 1 – Main Ideas

Three Levels of AI Users

- Level 1 – Dabblers: chase every new tool, collect accounts, have no system, and keep starting from scratch, so they stay busy but not effective.
- Level 2 – Skilled users: know how to prompt and get good results on individual tasks, but are reactive and don’t build bigger systems.
- Level 3 – Elite users: see many tools lightly, choose a few strategically, master them deeply and build systems that keep improving over time (“build the castle”).

Strategic Filtering and Staying Current

- Tier 3 users treat AI news like investment offers: most are ignored, a few get serious attention, based on real results not hype.
- They set a “news budget” (about 15–30 minutes per week) with 1–2 trusted sources instead of constantly scrolling for updates.
- For each promising tool they run a 30–60 minute trial on real tasks, then clearly decide to adopt (go deep), park (later) or reject (move on).

- Red flags = big marketing, weak examples, confusing setup, outputs needing huge fixes; green flags = clear time savings, good UX/docs and serious users reporting concrete gains.

Systems That Flow, Not Tool Chaos

- Dabblers stack separate tools for writing, research and organising, causing constant copy–paste, context loss and mental switching.
- Tier 3 users ask “Does this flow with my existing system?” and prefer AI that integrates where they already work (e.g. inside Docs or Notes).
- In a flowing system, research, outlining, drafting and editing all happen in one place, with AI remembering context and reducing friction, so improvements compound over time.

Evaluating AI Outputs and Task Stakes

- Tier 3 users judge every important AI output on five criteria: relevance, factual accuracy, completeness, clarity and whether it’s tailored or biased/generic.
- For high-value work they layer checks: cross-check sources, get an AI “second opinion”, use checklists and add human review.
- They classify tasks by stakes: low (brainstorms, quick summaries) where AI can run fast; medium (projects, important emails) where humans review; high (personal statements, decisions) where humans lead and AI only assists.
- They don’t trust AI blindly for everything or reject it completely; they adjust trust and verification depending on the risk.

Playbooks and Compounding Success

- A playbook is a written “AI recipe” for a repeated process, showing each step, what you do and what AI does, saved for reuse.
- You can build playbooks for research and writing, studying, projects, content creation or organising – any workflow you repeat often.
- Example steps: topic research, source review, outline, first draft, review and refine, each with a specific prompt or CO-STAR instruction for AI.

- Dabblers re-invent their process every time; Tier 3 users run and refine playbooks so each use makes the system better and their advantage grows.

Section 2 – Key Terms and Definitions

- Level 1 / Dabblers – AI users who chase every new tool, create many accounts and have no clear system, so they stay busy but don't improve much.
- Level 2 / Skilled Users – People who can prompt well and get good results on tasks but mainly use AI reactively and don't build larger workflows.
- Level 3 / Elite Users – Users who filter tools carefully, master a few deeply and build smooth systems where AI is woven into their daily work.
- Strategic Filtering – Choosing which tools and updates to try based on evidence and fit with your real tasks, not just hype.
- News Budget – A fixed small amount of time (e.g. 15–30 minutes per week) to scan AI news from 1–2 trusted sources without drowning in updates.
- 30–60 Minute Trial – A focused test where you use a new tool on real tasks, then decide to adopt, park or reject it.
- Red Flags – Warning signs that a tool isn't worth using, such as vague promises, poor examples, confusing setup or low-quality outputs.
- Green Flags – Positive signs that a tool is valuable, such as consistent time savings, clear design, good documentation and strong user results.
- System That Flows – A setup where AI is integrated into the tools you already use, so your work moves smoothly in one place with shared context.
- Tool Fragmentation – A messy situation where different tools handle separate pieces of work, forcing constant copying, pasting and context switching.
- Evaluation Criteria – The five standards Tier 3 users apply to outputs: relevance, factual accuracy, completeness, clarity and personal fit/bias.
- Human-in-the-Loop – A pattern where AI helps but a human reviews, checks facts and makes the final decisions.

- Low-Stakes Tasks – Activities like brainstorming or quick summaries where errors are not serious and AI can be trusted more.
- Medium-Stakes Tasks – Work like projects, important emails and research where AI helps but must be checked by a human.
- High-Stakes Tasks – Serious tasks (e.g. personal statements, decisions affecting others) where humans lead and AI only supports.
- Risk Classification – Sorting tasks into low, medium and high stakes to decide how much you should trust and verify AI.
- Playbook – A written process that shows how you and AI work together step-by-step on a repeated task.
- AI Integration Point – A specific step inside a workflow where AI is intentionally used to help (e.g. summarising sources, suggesting outlines).
- Compound Advantage – The growing benefit you get over time when you keep improving one system instead of restarting with new tools.
- Audit Your Setup – A review of which AI tools you have, which you really use and whether you have a system or just a random collection.

Section 3 – Short Summary

This lesson explains how to move from being a casual AI user to a “Tier 3” elite user in the top 1%. Instead of chasing every new tool, Tier 3 users apply strategic filtering, a small weekly news budget and focused trials to decide what is truly worth adopting. They avoid tool chaos by building systems that flow, where AI is integrated into the places they already work, so research, writing and editing all happen in one smooth process. For important outputs they have clear standards—relevance, accuracy, completeness, clarity and personal fit—and increase human checking as the stakes rise from low to high. A key idea is the AI playbook: a documented step-by-step workflow showing what you do and what AI does for any repeated task, which you refine over time. By auditing their setup, building filters and routines and running playbooks, Tier 3 users let their systems compound so they get further ahead while others stay stuck dabbling.

Section 4 – Quick Check Questions

1. What are the main differences between Level 1, Level 2 and Level 3 AI users?
2. What is a “news budget”, and how does it help you avoid drowning in AI updates?

3. Explain why systems that flow are better than using lots of separate AI tools for each small task.
 4. Name and briefly describe two of the five evaluation criteria Tier 3 users apply to important AI outputs.
 5. What is an AI playbook, and why does using playbooks give you a long-term advantage over dabbling?
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Lesson 7 – Your Future Strategy in the AI Era

Section 1 – Main Ideas

Big Picture – Where Value Is Moving

- AI is making some skills cheaper: pattern recognition, information lookup, standard analysis and routine problem-solving.
- Human-only skills are becoming more valuable: original thinking, judgment in messy situations, real relationships, navigating uncertainty, purposeful creativity and ethical reasoning.
- Economics idea: when something useful becomes scarce, its value rises – AI increases the supply of pattern skills, so deep human skills become the “rare flex”.
- People who use AI effectively already earn much more on average, and the gap is growing as roles and required skills change faster each year.

Your Edges Over AI

- Edge 1 – Human experience: you have lived failures, successes, fear, friendship and consequences, which grow intuition, empathy, judgment and wisdom that AI does not have.
- Edge 2 – Breaking patterns on purpose: AI follows and remixes patterns; humans can deliberately break them with meaning to create new products, art or ideas.
- Edge 3 – Feeling the weight of decisions: AI never feels guilt, responsibility or risk, but humans do, which matters when choices affect real people.

- Edge 4 – Caring about specific things: AI has no passions; your care for certain people, causes or crafts gives you direction, motivation and sharper insight.

“Bedside Manner” – Humans + AI Together

- An oncologist example shows that AI may diagnose better one day but will never have “bedside manner” – how you support someone in hard moments.
- Four human qualities that are hard to copy: courage, wisdom, humour and kindness/empathy; these are what people remember and pay for.
- Principle: let AI handle technical operations (drafts, analysis, admin) while humans own the relationship, trust and final judgment in jobs like law, teaching, medicine and sales.

Your Advantage Over Adults – Time and Plasticity

- Young people are “time billionaires”: time is your main currency and you have more of it now than you ever will again.
- Teenage brains have high neuroplasticity, so you can form new habits and skills faster than most adults and adapt quickly as tools and jobs change.

Strategic Mistakes to Avoid

- AI-dependency trap: moving from “AI helps me” to “I can’t do this without AI”, which weakens your own abilities instead of multiplying them.
- Information ≠ knowledge: AI gives instant facts, but real knowledge needs struggle, making connections, applying ideas and testing your understanding.
- Identity trap: building your identity on *how* you work (typing speed, memorising facts, old methods) instead of *what* you deliver (problem-solving, clear communication, building useful things).
- Balanced view: keep helpful old habits like deep reading and focused practice, but stay open to new tools so your “vehicle” can change while your destination (value) stays the same.

The Four Long-Term Assets

- Asset 1 – AI skills: understanding how AI works, when to use it (HEAD vs HEART), how to prompt and how to build systems puts you ahead of people who only “play” with tools.
- Asset 2 – Human skills: communication, emotional intelligence, collaboration, critical thinking and relationship building – essentially being a competent, moral, caring person.
- Asset 3 – Experience: actually building things in the real world (projects, writing, helping others) gives proof, judgment and resilience that pure reading never builds.
- Asset 4 – Flexibility: learning how you learn best and becoming comfortable with change so you can pick up new tools, roles and platforms quickly.

First Strategic Moves

- The lesson suggests practical next steps: honest self-assessment, checking where you rely on AI as a replacement, and writing a plan to grow each of the four assets.
- You should also audit your identity statements (“I’m someone who...”) and shift them from task-based (“I memorise facts”) to asset-based (“I solve complex problems and build things people use”).

Section 2 – Key Terms and Definitions

- Scarcity – When something useful is in short supply; in economics this usually makes it more valuable.
- Skill Obsolescence – When a skill becomes less useful or outdated because technology or jobs change.
- Half-Life of Skills – The idea that skills “expire” faster now, so what you learn may only stay current for a few years.
- Human Experience – Your lived moments of success, failure, risk and relationships that build intuition, empathy, judgment and wisdom.
- Bedside Manner – The caring, human way someone supports you when giving difficult news; used here as a symbol of human warmth that AI can’t copy.
- “Bedside Manner” Principle – Strategy where AI does the technical work and humans focus on trust, relationships and final decisions.

- Time Billionaire – A phrase meaning a young person who is “rich in time” and should treat time as their most precious currency.
- Neuroplasticity – The brain’s ability to change and form new connections; especially strong in teenagers, making learning faster.
- AI Dependency – Over-relying on AI so much that your own skills fade and you struggle to work without it.
- Multiplier vs Replacement – Using AI as a *multiplier* means you can do the task but AI boosts you; using it as a *replacement* means you can’t do the task without it.
- Information – Raw facts, data and text that AI can fetch instantly.
- Knowledge – Deeper understanding built by connecting information, applying it and learning from experience.
- Identity Trap – Building your sense of “who I am” on tools or methods (HOW) instead of on the deeper value you create (WHAT).
- Strategic Assets – Long-term strengths you can keep building: AI skills, human skills, real-world experience and flexibility.
- Builder Mindset – Choosing to create things (projects, writing, solutions) instead of only consuming content; summarised as “create > consume”.
- Flexibility / Adaptability – Being able to adjust smoothly when tools, roles or situations change, instead of freezing or resisting.
- Rapid Skill Acquisition – Having a clear system for learning new skills quickly and effectively as the world changes.
- Human Core – Your mix of character and people skills (communication, empathy, judgment, relationships) that stay valuable in any tech future.

Section 3 – Short Summary

This lesson is about positioning yourself to *win* in an AI-shaped future, not just survive it. As AI makes pattern-based tasks cheap, human-only abilities like judgment, deep relationships, purposeful creativity and ethical reasoning become rarer and more valuable. You also have built-in advantages: lived experience, the ability to break patterns with meaning, caring about specific things, and (as a young person) plenty of time and brain plasticity to learn fast. The main dangers are becoming dependent on AI, confusing information with real knowledge and

tying your identity to old methods instead of the value you create. To stay valuable, the lesson recommends building four long-term assets: strong AI skills, strong human skills, real-world experience and high flexibility. Finally, it offers practical “first moves” like honest self-assessment, checking where AI is a replacement, planning how to grow each asset and rewriting your identity around lasting strengths rather than tools.

Section 4 – Quick Check Questions

1. Why does AI make original thinking, judgment and relationships *more* valuable, not less?
2. Explain the “bedside manner” principle using one job (e.g. doctor, teacher, lawyer or salesperson).
3. What is the difference between using AI as a multiplier and using it as a replacement, and why does it matter?
4. Name and briefly describe two of the four strategic assets the lesson says you should build.
5. What is the identity trap, and how can you rewrite “I’m someone who...” statements to avoid it?